Section 5C. The Bootstrap

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Recap: Training Process

Recap: Training a parametric model

- We train our parametric models $f(\mathbf{x}_i; \theta)$ using a training dataset $\mathcal{D}_{\text{Tr}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$
- We assume each sample $(\mathbf{x}_i, y_i)$ is randomly drawn from a joint PDF f_{XY}
- Then, the estimated parameters

$$\widehat{\theta} = \arg\min_{\theta} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\mathbf{Tr}}} \left(y_i - f(\mathbf{x}_i; \theta)\right)^2$$

  is a random vector (it depends on data points that are random themselves)
- Also, our estimated output $f\left(\mathbf{x}; \widehat{\theta}\right)$ for a given input $\mathbf{x}$ is a random variable

Can we quantify the uncertainty in our estimated parameters and output?

An Unrealistic Approach

Before we explain the bootstrap, let us start with an **unrealistic approach** to quantify uncertainty

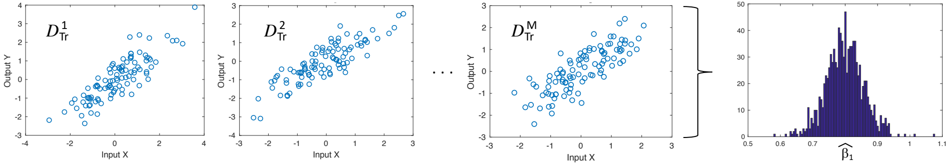
- Assume that, instead of a single training dataset, we have M (independent) training datasets, denoted by $\mathcal{D}_{\mathsf{Tr}}^{m}$ for $m = 1, \ldots, M$
- Using these datasets, we can compute M (independent) estimates of the unknown parameters, denoted by $\{\widehat{\theta}_m\}_{m=1}^{M}$
- Using these M estimates, we can visualize and study the uncertainty of our model; in other words, how robust they are to randomness in the data

Let us illustrate this idea numerically...

An Unrealistic Approach: Illustration

Example of a linear regression problem:

- We generate $M = 1000$ training datasets of a linear model with $\beta_1 = 0.8$ and $\beta_0 = 0$
- From these training datasets, we compute the estimates $\{\widehat{\beta}_1^m\}_{m=1}^{1000}$ (see histogram below)
- The mean of the histogram is 0.799 and the SD is 0.058; hence, the 95% CI for β_1 is $[0.683, 0.915]$



Notice that the approach above is *unrealistic* since, in practice, we do not have access to M (independent) training datasets

The Boostrap

Main Idea: Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, the bootstrap can generate M artificial datasets, as follows:

1. For $m = 1$ to M, do the following:
   1.1 Sample N random observations from $\mathcal{D}$ *with replacement*
   1.2 Include these N observations in an artificial dataset $\mathcal{D}_m^B$, called a bootstrap sample
   1.3 Train your model $f(\mathbf{x}; \theta)$ using the dataset $\mathcal{D}_m^B$, i.e.,

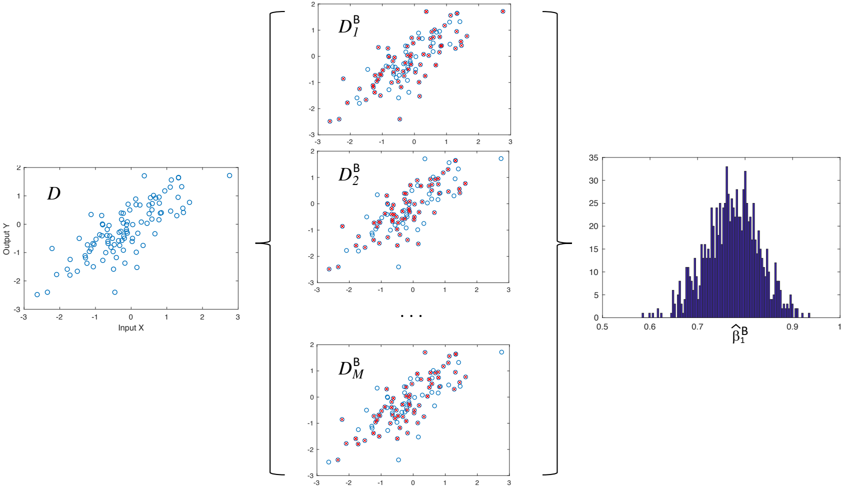
$$\widehat{\theta}_m^B = \arg\min_{\theta} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_m^B} (y_i - f(\mathbf{x}_i; \theta))^2$$

2. Estimate the standard deviation of the model coefficients using the formula

$$\mathrm{SD_B}\left(\widehat{\theta}\right) = \sqrt{\frac{1}{M-1} \sum_{m=1}^{M} \left(\widehat{\theta}_m^B - \overline{\widehat{\theta}^B}\right)^2}$$

   where $\overline{\widehat{\theta}^B} = \frac{1}{M} \sum_{m=1}^{M} \widehat{\theta}_m^B$ is the empirical mean

The Bootstrap: Visual Illustration



The Bootstrap: Numerical Example

Example of bootstrap in linear regression:

- We generate only one training dataset $\mathcal{D}$ of a univariate linear model with $\beta_1 = 0.8$
- Draw 1000 bootstrap samples $\{\mathcal{D}_m^B\}_{m=1}^M$. For each bootstrap sample, compute the estimate $\widehat{\beta}_1^m$ (see histogram at the right below). The mean of the histogram is 0.773 and the SD is 0.055

For comparison, we also plot the histogram of $\widehat{\beta}_1$ using 1000 independent realizations of the dataset (see histogram at the left below). The mean and SD are 0.799 and 0.058, respectively